

Valter Nunez

Software Engineer → Data Science | MSc Data Science, LSE (London)

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Summary

Software engineer moving into data science: a solid quantitative foundation (calculus, linear algebra, probability) and self-driven projects in **Bayesian inference** and natural-experiment design. Five years building data-intensive, event-driven backends for high-traffic e-commerce: schemas, telemetry, analytics pipelines. Now studying **MSc Data Science at LSE** in London (UK Graduate visa, 18 months, no sponsorship required); seeking Skilled Worker sponsorship to stay long-term.

Data Science Projects

- **Ballon d'Or attention study** — Bayesian two-stage (nomination/placement) decomposition in **PyMC** over ~210 player-seasons, isolating an unexplained media-attention residual after controlling for merit and team success; media pull on nomination is ~5× its pull on placement, robust across placebo and regression-to-the-mean checks. Ships as a D3 scrollytelling site + Quarto report.
- **World Cup 2026 momentum** — treats the tournament's mandatory hydration breaks as a natural experiment, using duration-matched stoppage comparisons + placebo tests; a live pipeline (local scrape → GitHub Actions → Pages) regenerates the report daily.
- **Insurance-claims regression** — implemented linear regression from scratch (variance, covariance, coefficients) and with scikit-learn, evaluated by RMSE on held-out Swedish auto-insurance data.

Experience

Backend Engineer

ZeBrands — high-traffic e-commerce

Sep 2021 – Mar 2026

Remote, Mexico

- Designed and built **OZ**, an event-driven platform that sustains sales through ERP outages, decoupling 83% of revenue-critical sales flows from synchronous ERP calls; designed the schemas and telemetry behind its reliability analytics (FastAPI, PostgreSQL, RabbitMQ, Redis); now live across ~195 stores (Mexico, expanding to Brazil), sustaining ~MX\$102M over the nine-day Hot Sale peak.
- Built multi-carrier shipping middleware with automated least-cost routing, cutting total shipping spend ~78%; migrated review analytics into a dedicated service for client-side insights.
- Maintained existing client-facing microservices and GraphQL data APIs (Frappe, TypeScript); stood up Grafana logging/observability (issue detection, performance dashboards).

Co-Founder & Developer

Laiter — software studio

2017 – 2019

Mexico

- Built Django web apps and the early usage-analytics dashboards clients ran decisions on.

Education

MSc Data Science

London School of Economics (LSE)

2026 – 2027

In progress, London

B.S. Computer Science & Technology

Tecnológico de Monterrey (ITESM)

2016 – 2021

Avg 90/100

Quantitative coursework: Calculus I–III, Linear Algebra, Probability & Statistics, Numerical Methods, Simulation. Semester abroad at RMIT Melbourne (Avg 94/100). IBM Data Science Professional Certificate (in progress).

Technical Skills

Data Science: Python, pandas, NumPy, scikit-learn, PyMC, Bayesian inference, statistics, matplotlib/D3, Quarto

AI / LLM: LLM integration (Gemini) with metering & structured output; agent-orchestrated development

Data & Databases: SQL, PostgreSQL, Redis, MongoDB, RabbitMQ

Engineering: FastAPI, Django, TypeScript, Docker, Git, GitHub Actions

Languages spoken: Spanish (Native), English (C2), German (A2), Italian (A1)